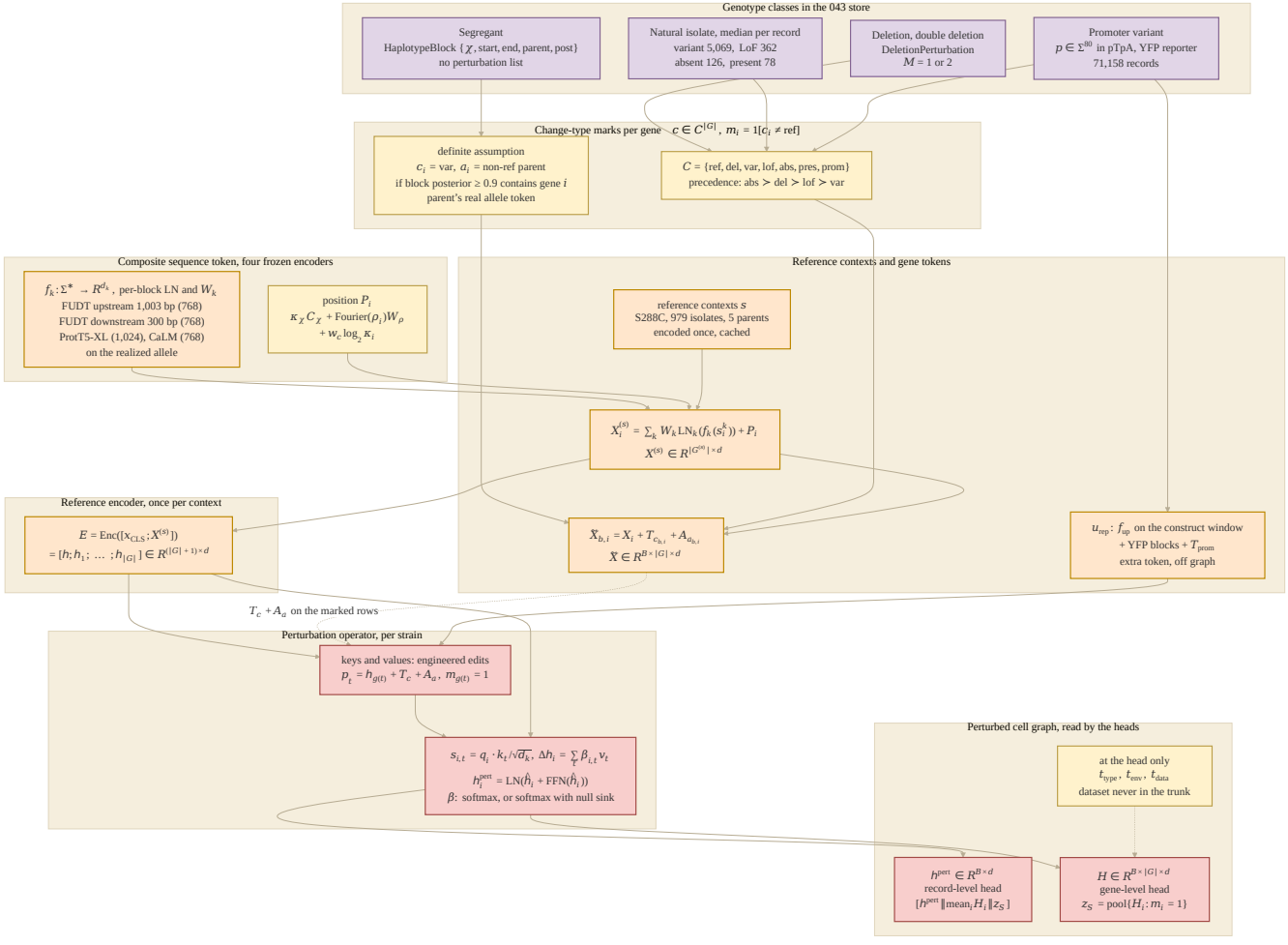

Genotype classes in the 043 store
Segregant
HaplotypeBlock {χ, start, end, parent, post}
no perturbation list
Natural isolate, median per record
variant 5,069, LoF 362
absent 126, present 78
Deletion, double deletion
DeletionPerturbation
M = 1 or 2
Promoter variant
p ∈ Σ^80 in pTpA, YFP reporter
71,158 records
Change-type marks per gene c ∈ C^|G| , m_i = 1[c_i ≠ ref]
definite assumption
c_i = var, a_i = non-ref parent
if block posterior ≥ 0.9 contains gene i
parent's real allele token
C = {ref, del, var, lof, abs, pres, prom}
precedence: abs ≻ del ≻ lof ≻ var
Composite sequence token, four frozen encoders
f_k : Σ* → R^{d_k} , per-block LN and W_k
FUDT upstream 1,003 bp (768)
FUDT downstream 300 bp (768)
ProtT5-XL (1,024), CaLM (768)
on the realized allele
position P_i
κ_χ C_χ + Fourier(ρ_i)W_ρ
+ w_c log_2 κ_i
Reference contexts and gene tokens
reference contexts S
S288C, 979 isolates, 5 parents
encoded once, cached
X_i^(s) = Σ_k W_k LN_k(f_k(s_i^k)) + P_i
X^(s) ∈ R^{|G^ref| × d}
Reference encoder, once per context
E = Enc([x_CLS; X^(s)])
= [h; h_1; … ; h_|G|] ∈ R^{(|G|+1)×d}
X̃_{b,i} = X_i + T_{c_{b,i}} + A_{a_{b,i}}
X̃ ∈ R^{B×|G|×d}
u_rep : f_up on the construct window
+ YFP blocks + T_prom
extra token, off graph
T_c + A_a on the marked rows
Perturbation operator, per strain
keys and values: engineered edits
p_t = h_{g(t)} + T_c + A_a , m_{g(t)} = 1
s_{i,t} = q_i · k_t / √d_k , Δh_i = Σ_t β_{i,t} v_t
h_i^pert = LN(ĥ_i + FFN(ĥ_i))
β: softmax, or softmax with null sink
Perturbed cell graph, read by the heads
at the head only
t_type , t_env , t_data
dataset never in the trunk
h^pert ∈ R^{B×d}
record-level head
[h^pert ‖ mean_i H_i ‖ z_S]
H ∈ R^{B×|G|×d}
gene-level head
z_S = pool{H_i : m_i = 1}